

ARIA

Agentic Real-Estate Intelligence Advisor

A multi-agent machine-learning & generative-AI system that turns 135,051 public Airbnb listings across Paris and Athens into explainable pricing, risk, demand, and compliance recommendations — one defensible answer, not four disconnected dashboards.

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€7.7M

total annual opportunity & exposure identified across pricing, risk, demand & compliance — Athens

135,051

listings analysed across Paris & Athens, with 96 engineered columns

865

listings simultaneously underpriced AND high-risk — the priority action list

5

specialist ML / GenAI agents orchestrated by one LangGraph supervisor

01 Objective

Hosts don't know if they're leaving revenue on the table. Investors can't easily compare neighbourhoods. Regulators lack a way to find non-compliant supply. **ARIA closes this gap** with one explainable, multi-agent system.

THE PREMISE

Pricing, risk, demand & compliance combined into one defensible recommendation — mirroring how a real-world advisory engagement is actually delivered.

02 Data Sources

Four independent public sources combine into one master dataset — **135,051 listings × 96 columns** across two structurally different markets.

Maven Analytics · Paris 2021	63,520
Inside Airbnb · Paris 2025	57,289
Inside Airbnb · Athens 2025	14,242

41-step pipeline: encoding fixes, log-price transform, Haversine distance zones, VADER sentiment (bias-corrected), and routing flags that keep estimated prices out of training.

2.8×

Key EDA finding. Athens's distance-to-centre price effect is 2.8× stronger than Paris's — the quantified justification for training two separate city models, not one pooled model.

03 Methodology & Stack

Six validated ML/data phases, one orchestration layer, one production interface — every model seeded (**RANDOM_STATE = 42**), tuned with 100-trial Optuna search, validated on a locked 80/20 holdout, and explained with SHAP.



04 Pricing

XGBoost predicts fair nightly price and flags listings priced well below model fair value.

0.588

Paris R²

0.676

Athens R²

€29

MAE

Why trees, not a neural net? Price is non-linear & interaction-heavy; trees capture it without hand-built terms and stay explainable to a KPMG audience. SHAP shows genuinely different drivers per city.

BUSINESS OUTPUT

2,945 Athens listings underpriced >€15 → an estimated **€4.8M/year** foregone revenue.

05 Risk

LightGBM flags Athens listings showing early host-vulnerability signals — a probability, not a verdict.

0.829

AUC-ROC

0.886

Avg. precision

0.166

Brier score

Trusted AI over a vanity metric. A first model scored an implausible 0.9995 AUC. SHAP caught the leak, we fixed it, and report the honest 0.8288 here.

BUSINESS OUTPUT

4,695 high-risk listings — 865 also underpriced: the single best action list, worth **€1.43M**.

06 Demand

Prophet forecasts 12-month occupied-nights per city with 90% confidence intervals — giving pricing & risk forward-looking context.

— Athens — Paris



Both cities signal a growing demand trend into 2026; the band is the Athens 90% CI, widening with horizon.

07 Compliance

A ChromaDB RAG agent retrieves the specific applicable article from the Athens AMA & Loi Le Meur corpus per listing — not just a flag.

137

unlicensed listings

27

high-risk overlap

€991k

revenue at risk

SCOPE

Analyst-triage support for advisory teams — explicitly never a final legal determination or enforcement trigger.

08 Orchestration

A 9-node LangGraph StateGraph routes each question to the right specialist, synthesises an answer, and pauses at a **human-in-the-loop checkpoint** — a Trusted AI human-in-the-loop requirement, enforced in code.

100%

routing accuracy

17

state fields tracked

≥97

answer-quality score

Deterministic models compute every number. The language model only narrates — never the source of a number. A guaranteed fallback keeps every answer sourced even if the model call fails.

Validated

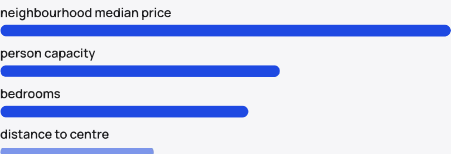
Every model meets or exceeds its pre-declared target on a locked holdout.

EDA coverage	99%
XGBoost R ² (Athens)	0.676
LightGBM AUC	0.829
RAG routing accuracy	100%
Answer quality	≥97

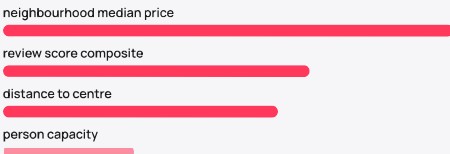
Why We Model Each City Separately

Cross-city SHAP proves the markets are driven by **genuinely different mechanics** — not just different data quality. Neighbourhood market context ranks #1 in both; everything below diverges.

PARIS — SIZE & CAPACITY



ATHENS — QUALITY & LOCATION



What Makes ARIA Different

- > **Not one model** — a team of five. Pricing, risk, demand & compliance specialists behind one agentic orchestrator, not a single dashboard.
- > **Grounded GenAI.** Models compute every number; the language model only narrates — so no figure is ever hallucinated.
- > **Regulation-aware.** The RAG agent links each listing to the specific law it may breach — not just a yes/no flag.
- > **Trusted-AI by discipline.** It caught and corrected its own data leakage rather than reporting a too-good score.

€ Business Impact — Athens Market

€4.8M

Pricing gap

€1.4M

Host-risk revenue

€1.0M

Demand growth

€0.5M

Compliance exposure

€7.7M

Total opportunity